IRIS

Uvic Hacks Startup Hackathon 2026

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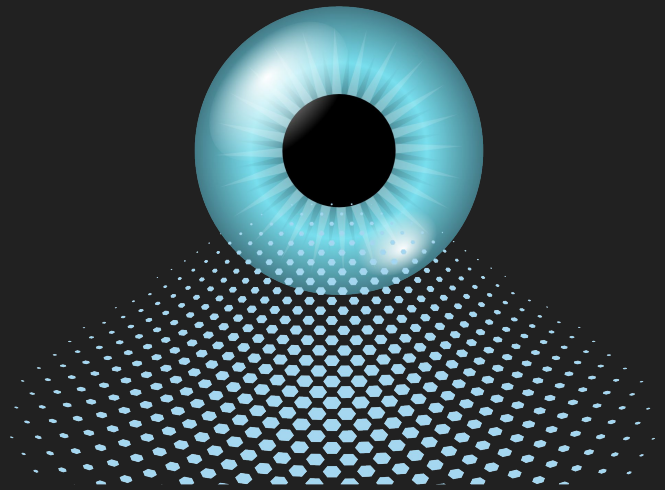
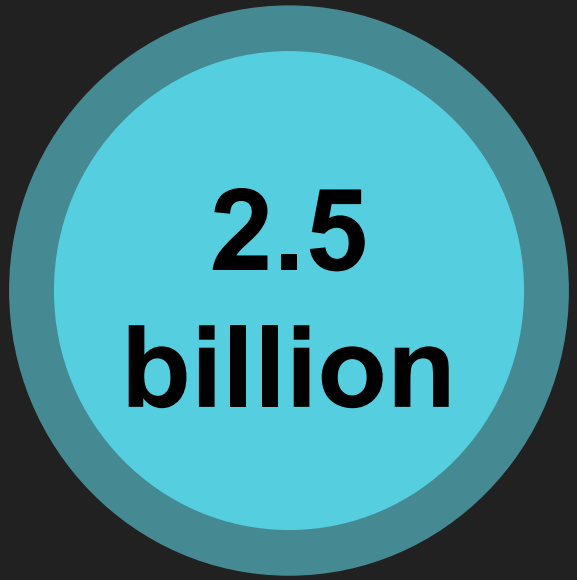


The Problem



People need assistive products, such as wheelchairs, hearing aids, or apps that support communication and cognition.

Our Solution

This project provides a low-cost app that works on standard smartphones and tablets.

It helps users gain more independence by removing the need for bulky or specialized hardware.

United Nations SDG’s

#3 Good Health & Well-Being

#10 Reduced Inequalities

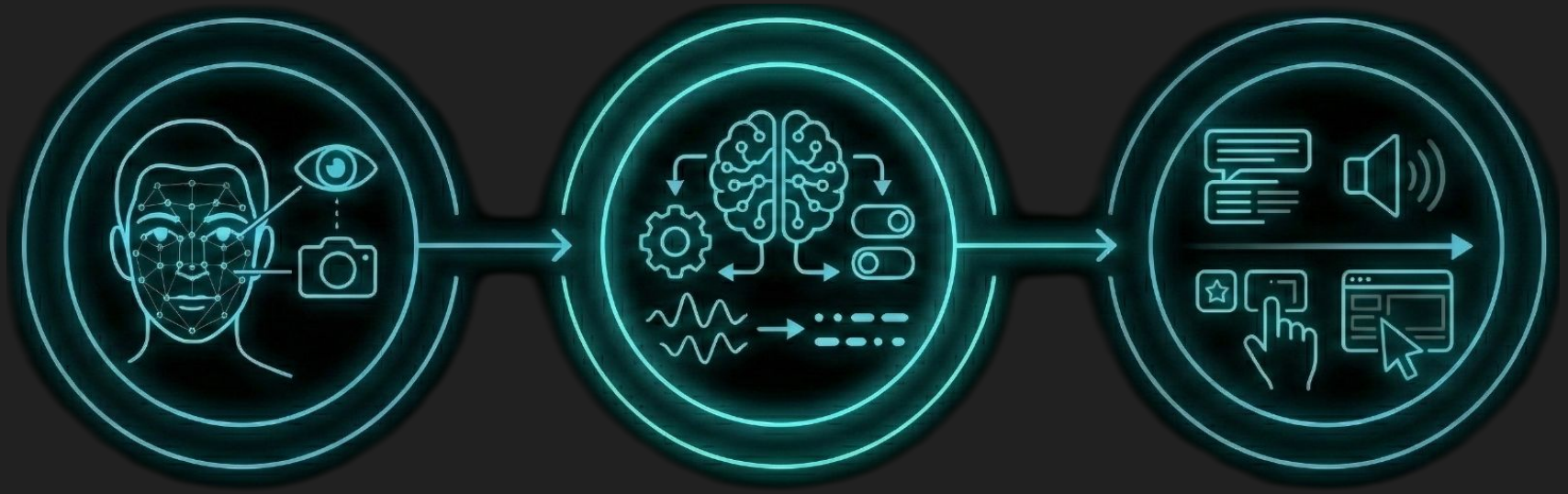
#4 Quality Education

Vision & purpose

Inspired by witnessing the struggle of a classmate with Rett Syndrome.

Iris was developed to dismantle the cost and hardware barriers of modern assistive technology.

How It Works



Visual Capture
Maps 68 facial landmarks in real-time using your existing camera. No specialized sensors or expensive hardware required.

Neural Processing
Our engine distinguishes between natural involuntary movement and intentional long/short triggers to form Morse patterns.

Dynamic Output
Patterns are decoded into text, natural speech, or system navigation, enabling total digital independence.

Implementation

Front end: HTML, CSS, JavaScript

The morse page shows the camera in an image and talks to the server with simple requests; spoken feedback uses the browser's built-in text-to-speech.

Data base: SQLite

Stores users and roles. No separate database server; the app creates and uses the file itself.

Backend / APIs: Python with Flask.

Handles pages, login/signup (with secure passwords), and the Morse feature.

The morse part uses the webcam and face-tracking to detect blinks, turns them into Morse.

Business model

14-Day full-access **free trial**:

- Includes full access to the vision model (morse-to-text) and Text-to-Speech (TTS) engine.

Subscription Model:

- Individual (**$100+/mo**): Unlimited morse-to-text and early access to new features.

Zero-hardware advantage: No upfront cost - immediate ROI for families by using any existing iPad, Tablet, or Laptop.

Live Demo

Marketing/GTM strategy

Targeting Speech-Language Pathologists (SLPs) and Occupational Therapists at rehab centers.

Strategic Partnerships:

- **Hospital & rehab pilot programs**: Deploying the software in ICU and long-term care units for communication.
- **Patient advocacy groups**: Partner with the ALS Society of Canada, Muscular Dystrophy Canada, and MS foundations for awareness and "vetting."
- **Assistive tech influencers**: Engaging with "Adaptive Tech" YouTube/TikTok creators who review tools for the disabled community.

What’s Next?

Multi-Interface Support:

This app will support different forms of communication and ‘typing’

Multilingual Support

The app will support multiple languages and translations

Thank you!

Competitors

Feature	IRIS	Eyegaze Inc. (Edge)	Tobii Dynavox (I-Series)	Control Bionics (NeuroNode)
Input Method	Blink-to-Morse	Gaze Selection (Staring)	Gaze, touch, & switch	EMG (Muscle twitch) + gaze
Morse Based?	Yes (Primary)	No (On-screen keyboard)	No (On-screen keyboard)	No (On-screen keyboard)
Hardware	Any Webcam/Phone	Specialized IR Tablet/Bar	Rugged medical tablet	Wearable sensor + tablet
Paraplegic Ready?	Yes (Extreme mobility)	Yes (Gold standard)	Yes	Yes (Less fatiguing)
Estimated Cost	$100+	$10,000+	$12,000 - $15,000	$14,000+

A Story of Independence

The Challenge

- Sarah has limited speech and hand mobility.
- Traditional tools are expensive and bulky.
- Communication is often tied to a single desk.

The Impact

- **Portability**: Works in parks, restaurants, or on the go.
- **Independence**: No specialized hardware or technicians required.
- **Connection**: Sarah joins conversations simply by looking at her screen.

Context

Rapid Onset

- Guillain-Barre syndrome
- Post Stroke Motor Impairment

Slow Onset

- ALS (Amyotrophic lateral sclerosis)
- Parkinson’s Disease

References

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2. World Health Organization, "Disability," Mar. 7, 2023. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/disability-and-health. [Accessed: Feb. 28, 2026].
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